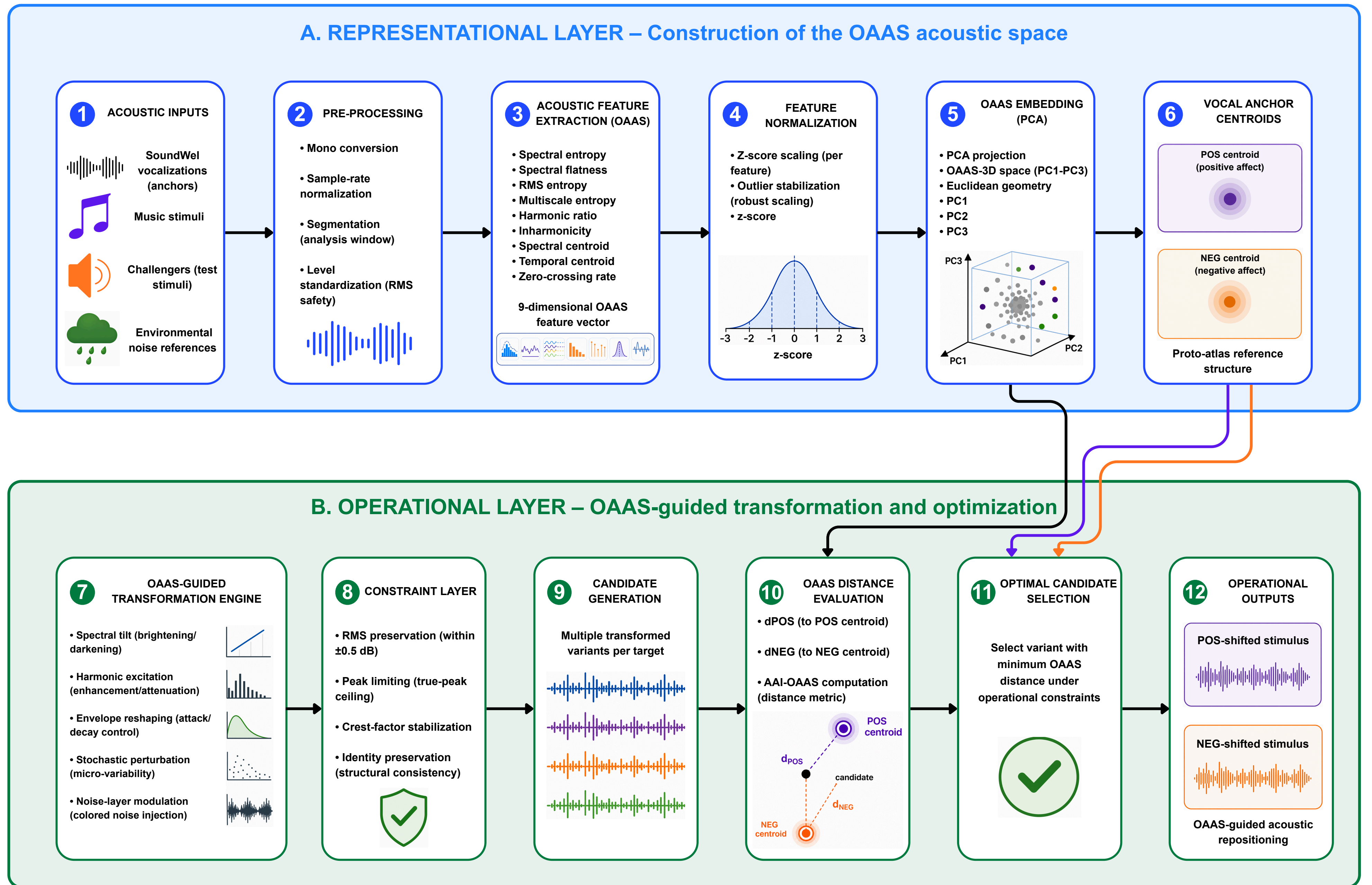


Supplementary Methods Figure SM1
Computational and operational architecture of the OAAS framework



Caption. Acoustic inputs (vocalizations, musical stimuli, challengers, and environmental references) were preprocessed and projected into the OAAS representational space through multidimensional acoustic feature extraction, z-score normalization, and PCA embedding. The operational PCA model used a reduced 9-dimensional feature vector comprising spectral entropy statistics, spectral flatness statistics, RMS temporal entropy, multiscale envelope entropy, and harmonic ratio descriptors. Vocal anchor centroids defined operational affective targets guiding the constrained transformation engine. Iterative transformations were performed under RMS-preservation and peak-safety constraints, enabling directional OAAS repositioning while preserving global acoustic identity and level stability.